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## EDUCATION

### University of Toronto

*PhD Candidate, Electrical Engineering/Electromagnetics*

*Toronto, Canada*

*Sep 2021 — Present*

- Using numeric methods, optimization, and finite-element simulation tools to design and optimize relevant antenna structures, e.g. COMSOL, HFSS, and the method of moments (MoM). Designs are fabricated in-lab using laser printed circuit board (PCB) prototyping machine. Experimental verification is performed using relevant RF measurement setups.

### University of Toronto

*Master of Applied Science (MAsc), Electrical Engineering/Electromagnetics*

*Toronto, Canada*

*Sep 2019 — Sep 2021*

- Received Honorable Mention, 2021 IEEE AP-S/URSI Student Paper Competition.

### University of Toronto

*Honors Bachelors of Science (BSc Hons), Physics & Mathematics Double Major*

*Toronto, Canada*

*Sep 2015 — Jun 2019*

- Dean's List Award 2016-2018.
- Received New College Council In-Course Scholarship 2016.

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## PROFESSIONAL EXPERIENCE

### Research & Development Intern

*Ansys (part of Synopsys): Electronics, Simulation, and Optics Unit*

*Remote, Canada*

*May 2025 - Apr 2026*

- **Developed, implemented, and tested** methods for enhancing the prediction accuracy of surrogate thermal models for integrated circuit (IC) packages in Ansys Icepak, using a **physics-informed data sampling approach**. Performed detailed study of effects with respect to dataset size, data quality, convergence, etc. **Achieved 60-80% error reduction** for tested IC packages compared to conventional surrogate modeling techniques.
- **Created general-purpose data sampling workflow** for surrogate model training based on observed package sensor data, allowing accurate inference of underlying physics behavior. **Achieved 60-80% error reduction** for tested IC package.
- **Developed Bayesian inference analysis pipeline** to create robust, adaptive data sampling method capable of responding in real-time to live IC sensor data.
- Authored relevant academic publication on project (in progress). US patent application in progress: "**Physics-Informed Adaptive Model Calibration Using Operating Regime Detection System for Multiphysics Simulations.**"
- Recognized with **ONEAnsys company internal award** for technical contributions.
- Leveraged **PyCharm** and **modular concepts** for code development.

### Researcher, Computational Physics/Atomistic Simulation

*SickKids Research Institute*

*Toronto, Canada*

*Jan 2015 — Sep 2019*

- **Numerically simulated** atomic interactions and chemical structures using molecular dynamics on the **SciNet supercomputer** cluster. Analyzed, preprocessed, and visualized **large-scale datasets** using **Python**.
- **Identified and categorized distinct preferential binding modes** consistent with previous literature on amyloid-beta oligomer behavior. Characterized binding propensities using statistical methods.

### Research Assistant, Computational Solid State Physics

*University of Toronto: Department of Physics*

*Toronto, Canada*

*May 2018— Aug 2018*

- **Constructed new reduced tight-binding model of nickel (II) iodide** by calculated tight-binding parameters for electron-hopping model. Performed numeric simulation and data analysis of electron-hopping models using Python via **Monte Carlo simulation** and **statistical methods**.
- **Generated datasets from density functional theory (DFT) using HPC tools** on SciNet supercomputer cluster.
- **Used Python to calculate energy band structure from new tight-binding model** and confirm agreement with experimental data from literature. Identified distinct magnetic ordering cases via simulated annealing using Monte-Carlo methods.
- Received **NSERC HEATER scholarship** in thermoelectrics training and research.

### Research Assistant, Data Quality and Output

*SNOLAB: SNO+ Experiment Group*

*Sudbury, Canada*

*May 2017— Aug 2017*

- **Developed and debugged C++ scripts for data quality checks** on SNO+ detector output ROOT and relevant physics libraries.
- **Assembled and tested calibration apparatus** for SNO+ detector per clean room protocols. Assisted with equipment assembly in ultra-clean underground facility at 2 km depth.
- Received **NSERC summer research scholarship**.

## Research Assistant, Atomic Physics

University of Toronto: Department of Physics

Toronto, Canada

May 2016— Jun 2016

- **Used RF equipment to test a small solid-state NMR apparatus** to measure precession frequencies of magnetic dipole moments in Pb-207 nuclei, in the context of searching for violations in time reversal symmetry.
- **Designed and built RF filters** and relevant RF setup for experiment.
- Received departmental **Summer Undergraduate Research Fellowship (SURF)**.

## TECHNICAL SKILLS

|                                    |                                                                                                                                                                                                                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>      | Python, MATLAB, C++, Java                                                                                                                                                                                                                                                                                     |
| <b>Simulation Tools:</b>           | COMSOL Multiphysics, Ansys HFSS, Ansys Icepak                                                                                                                                                                                                                                                                 |
| <b>Modeling/Numerical Methods:</b> | Finite-element method (FEM), finite-difference time-domain method (FDTD), integral equation methods/method of moments (MoM), computational electromagnetics (CEM), thermal simulation, multiphysics simulation, reduced order modeling, surrogate modeling, numeric optimization, computer-aided design (CAD) |
| <b>AI/ML Tools:</b>                | Tensorflow, Keras, PyTorch, Claude Code                                                                                                                                                                                                                                                                       |
| <b>Software Tools:</b>             | Anaconda, Jupyter Notebook, PyCharm, VSCode, Git, Linux, Bash, high performance computing (HPC), LaTeX, Microsoft Office                                                                                                                                                                                      |
| <b>Analog Design/EDA:</b>          | Cadence Innovus, Cadence Genus, RTL, TCL, Keysight ADS, KiCad                                                                                                                                                                                                                                                 |
| <b>RF/Electronics:</b>             | Radiofrequency (RF) design, RF equipment (oscilloscope, network analyzer, spectrum analyzer, waveform generator), antenna design, antenna measurement, printed circuit board (PCB) prototyping, digital integrated circuits (IC), laser PCB etching                                                           |
| <b>Manufacturing/Other:</b>        | 3-D printing, composite materials, machining                                                                                                                                                                                                                                                                  |
| <b>Subject Areas:</b>              | Electromagnetics, physics, metamaterials, wireless power                                                                                                                                                                                                                                                      |

## ACADEMIC PUBLICATIONS AND PATENTS

### Physics-Informed Adaptive Model Calibration Using Operating Regime Detection System for Multiphysics Simulations

U.S. Patent Application, co-inventor, assigned to Synopsys Inc. (in progress, 2026)

### Wireless Power Transfer Using a Cavity-Enclosed Impedance Metawire Structure, [link](#)

2026 IEEE International Symposium on Antennas and Propagation, Detroit, United States

### An Aperiodic Sub-Wavelength Dipole Structure for Enhancing Near-Field Wireless Power Transfer, [link](#)

IEEE Antennas and Wireless Propagation Letters (AWPL), Vol. 23, No. 6, July 2024

### A Shifted-Beam Method for Near-Field Wireless Power Transfer using Parasitic Arrays, [link](#)

2023 IEEE International Symposium on Antennas and Propagation, Portland, United States

### Shifted-Beam Array Coil for Highly Focal Transcranial Magnetic Stimulation, [link](#)

2021 IEEE International Symposium on Antennas and Propagation, Singapore

## ORGANISATIONS AND VOLUNTEERING

### Vice Chair, University of Toronto IEEE Antenna and Propagation Society (AP-S) Student Chapter

Oct 2023 — Dec 2025

- Hosted social and academic events for University of Toronto IEEE AP-S student chapter. Elected position.

### Team Lead & Composites Fabricator, Rocketry Division, University of Toronto Aerospace Team (UTAT)

Sep 2015 — Aug 2021

- Led a team of more than 10 members for rocketry airframe subsystem and solid-fuel rocket program (2019-2021).
- Used machine tools and composite manufacturing techniques to fabricate carbon fiber airframes for low-altitude rockets.
- Performed physical compression testing to verify performance. Coordinated design and construction of solid-fuel test rockets.

## OTHER

- Canadian citizen.
- Languages: English (fluent), French (basic proficiency).